

Day 05 - Git Rebase, Reset & Reflog

Objectives

Understand Rebase, Interactive Rebase, Reset and Reflog with production workflows.

Git Rebase

`git rebase main`

Creates a linear history by replaying commits. New commit SHAs are created. Avoid rebasing shared branches.

Interactive Rebase

`git rebase -i HEAD~4`

pick, squash, reword, edit, drop.

Git Reset

--soft: move HEAD, keep staged.

--mixed: move HEAD, unstage changes.

--hard: move HEAD and discard local changes.

Git Reflog

`git reflog` records HEAD movements.

Recover with: `git reset --hard HEAD@{1}`

Production Workflow

`git fetch origin` → `git rebase origin/main` → resolve conflicts → `git rebase --continue` → `git push --force-with-lease`

Interview Questions

Merge vs Rebase? Why SHA changes? Difference between soft/mixed/hard reset? What is Reflog?

Key Takeaways

Rebase keeps history clean. Reset changes HEAD behavior. Reflog recovers lost commits.